

Pranav Hirawat

AI engineering — LLM integration, applied ML, and the systems that keep them honest.

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SKILLS

Languages	Python, Go, Swift, SQL
AI / ML	LLM integration (Ollama), prompt-injection mitigation, provider abstraction and fallback, scikit-learn, NLP, TF-IDF, model evaluation (precision / recall / F1, confusion matrix)
Systems	Redis (streams, consumer groups, TTL policy), event-driven pipelines, agent loops
Practice	pytest, property-based testing, Git, SwiftUI, Streamlit

PROJECTS

Agentic Control Plane — Intelligent API Security Gateway

Jan 2026 – Present

PES University · four-person project · sole author of the Python control plane

- **Integrated an LLM (Ollama) to write incident summaries, sequenced strictly after enforcement** — a prompt-injected summary can mislead a human reader but cannot unblock an attacker. Narration is optional: a null provider keeps the system fully functional with no model present.
- Built a 30-second agent loop — observe, correlate, decide, act, explain — that reads attack evidence from a Redis stream and writes enforcement policy back with a TTL, so every decision expires on its own rather than needing revocation.
- Designed union-find correlation over shared traits (endpoint, User-Agent, subnet, timing) that groups scattered IPs into one campaign, so an attacker rotating addresses is still recognised as a single actor.
- Mapped severity and confidence onto an action ladder (monitor → throttle → temporary block → escalate); enforcement that fails promotes a campaign rather than repeating itself.
- **249 tests against 3,925 lines of source (0.84:1 test-to-code)**, testing properties rather than paths — including that narration failures can never fail a cycle and that clean traffic produces zero detections. One runtime dependency; the rest is standard library.

Fake Review Detector

Mar – Apr 2026

PES University

- Built an NLP classifier distinguishing computer-generated product reviews from genuine ones, over a cleaned, stopword-stripped, TF-IDF-vectorised corpus.
- Compared logistic regression against naive Bayes, reaching **~89% accuracy**, and evaluated with a confusion matrix and precision / recall / F1 rather than accuracy alone — on a task where a false accusation costs more than a miss.
- Deployed as an interactive Streamlit app scoring pasted reviews in real time.
- Live: fake-review-detector-pranav.streamlit.app · Code: github.com/Pranav-Hirawat/Fake-review-detector

Helivor — Disaster Management iOS App

Oct 2025

PES University

- Built a SwiftUI app that keeps its intelligence **on-device**, so guidance and local weather conditions stay available when disaster has taken out connectivity — the moment a server round trip is least likely to succeed.
- Code: github.com/Pranav-Hirawat/disastermanagement

EDUCATION

Bachelor of Computer Applications (BCA)

Expected 2027

PES University, Bengaluru